Page 1: Introduction to AI

Artificial Intelligence (AI) is the simulation of human intelligence in machines.

It enables computers to perform tasks such as reasoning, learning, and problem-solving.

AI is widely used in industries like healthcare, finance, and automation.

Page 2: Machine Learning

Machine Learning (ML) is a subset of AI that allows systems to learn from data.

Instead of being explicitly programmed, ML models improve through experience.

Examples include recommendation systems and fraud detection.

Page 3: Retrieval-Augmented Generation (RAG)

RAG combines retrieval of relevant information with generation using LLMs.

It helps reduce hallucination and improves factual accuracy.

RAG is widely used in enterprise chatbots and knowledge systems.

Page 4: Embeddings and Vector Databases

Embeddings convert text into numerical vectors.

These vectors help measure similarity between pieces of text.

Vector databases like FAISS store and retrieve these embeddings efficiently.

Page 5: Real-World Applications

RAG systems are used in:

- Customer support chatbots
- Legal document analysis
- Healthcare assistants
- Internal company knowledge search

They improve productivity and decision-making.